FREQUENCY RESPONSE

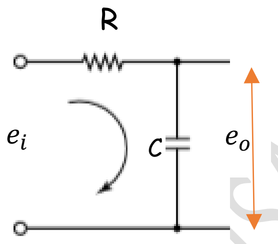
The steady state response of a system to a sinusoidal input is known as frequency response. A sinusoidal input to a linear sytem gives a sinusoidal out put of the same frequency. But the amplitude and phase of the output may differ from that of input. These differences will be functions of frequency.

Let us consider a RC network as shown. We can write,

$e_i = iR + \frac{1}{C}\int i\,dt$ and $e_o = \frac{1}{C}\int i\,dt$. Taking Laplace transform

$$E_i(s) = R\,I(s) + \frac{1}{C}\frac{I(s)}{s} = I(s)\left(R + \frac{1}{Cs}\right)$$

$$E_o(s) = \frac{1}{C}\frac{I(s)}{s}$$



The transfer function of the system

$$G(s) = \frac{E_o(s)}{E_i(s)} = \frac{\frac{1}{Cs}I(s)}{I(s)\left(R + \frac{1}{Cs}\right)} = \frac{1}{1 + RC\,s}$$

$$= \frac{1}{1 + Ts}$$

This is a first order system with time constant $T = RC$. This can also be written as

$$G(s) = \frac{1}{T(s + \frac{1}{T})} = \frac{a}{s + a}$$

The output $E_o(s) = G(s)E_i(s)$

Let the input voltage be $e_i = A\sin\omega t$

$$E_i(s) = \frac{A\omega}{s^2 + \omega^2}$$

$$E_o(s) = \frac{a}{s + a}\,\frac{A\omega}{s^2 + \omega^2} = \frac{B}{s + a} + \frac{Ps + Q}{s^2 + \omega^2}$$

$$A a\omega = B(s^2 + \omega^2) + (Ps + Q)(s + a)$$

Put $s = -a$ >>> $B = \frac{Aa\omega}{(a^2+\omega^2)}$

Equating coefficients of s^2 >>> $P = \frac{-Aa\omega}{(a^2+\omega^2)}$

Equating coefficents of s >>> $Q = \frac{+Aa^2\omega}{(a^2+\omega^2)}$

$$E_o(s) = \frac{\frac{Aa\omega}{(a^2+\omega^2)}}{s + a} + \frac{\frac{-Aa\omega}{(a^2+\omega^2)}s + \frac{Aa^2\omega}{(a^2+\omega^2)}}{s^2 + \omega^2} = \frac{Aa\omega}{(a^2+\omega^2)}\left[\frac{1}{s + a}\right] - \frac{Aa\omega}{(a^2+\omega^2)}\left[\frac{s}{s^2 + \omega^2}\right] + \frac{Aa^2}{(a^2+\omega^2)}\left[\frac{\omega}{s^2 + \omega^2}\right]$$

Taking inverse Laplace transform

$$e_o(t) = \frac{Aa\omega}{(a^2+\omega^2)}e^{-at} - \frac{Aa\omega}{(a^2+\omega^2)}\cos\omega t + \frac{Aa^2}{(a^2+\omega^2)}\sin\omega t$$

At steady state $e^{-at} \to 0$ as $t \to \infty$

$$e_o(t) = -\frac{Aa\omega}{(a^2+\omega^2)}\cos\omega t + \frac{Aa^2}{(a^2+\omega^2)}\sin\omega t$$

Put $\frac{-Aa\omega}{(a^2+\omega^2)} = X\sin\theta$ and $\frac{Aa^2}{(a^2+\omega^2)} = X\cos\theta$ sothat $\left[\frac{Aa\omega}{(a^2+\omega^2)}\right]^2 + \left[\frac{Aa^2}{(a^2+\omega^2)}\right]^2 = X^2$ and $\tan\theta = -\frac{\omega}{a}$

$$X = \sqrt{\left[\frac{Aa\omega}{(a^2+\omega^2)}\right]^2 + \left[\frac{Aa^2}{(a^2+\omega^2)}\right]^2} = \frac{Aa}{\sqrt{(a^2+\omega^2)}}$$

$$e_o(t) = X\sin\theta\,\cos\omega t + X\cos\theta\,\sin\omega t = X\sin(\omega t + \theta) = \frac{Aa}{\sqrt{(a^2+\omega^2)}}\sin(\omega t + \theta) = \frac{A}{\sqrt{1 + \omega^2T^2}}\sin(\omega t + \theta)$$

This is the frequency response of the given first order system. The magnitude of the frequency response is $\frac{A}{\sqrt{1+\omega^2T^2}}$ and phase angle $\theta = \tan^{-1}\left(-\frac{\omega}{a}\right) = \tan^{-1}(-\omega T) = -\tan^{-1}(\omega T)$

We can get the same results if we replace s in $G(s)$ by the complex frequency $j\omega$

$G(s) = \frac{1}{1 + Ts}$	$G(j\omega) = \frac{1}{1 + j\omega T}$	$G(j\omega) = \frac{1 - j\omega T}{1 + \omega^2T^2}$	$G(j\omega) = \frac{1}{1 + \omega^2T^2} - j\frac{\omega T}{1 + \omega^2T^2}$

$$M = |G(j\omega)| = \frac{1}{\sqrt{1 + \omega^2T^2}} \quad ; \qquad \theta = \angle G(j\omega) = \tan^{-1}(-\omega T)$$

The frequency response $e_o(t) = A.|G(j\omega)|\ \sin(\omega t + < G(j\omega))$

Magnitude of the output $M_o = M_i.M$ and $\emptyset_0 = \emptyset_i + \theta$

$G(j\omega)$ is known as the sinusoidal transfer function. The sinusoidal transfer function, a complex function of frequency ω, is characterised by its magnitude and phase angle, with frequency as a parameter. There are three commonly used represetations of sinusoidal transfer functions

1. Bode diagram or logarithmic plot
2. Nyquist plot or polar plot
3. Nichols plot or log magnitude vs phase plot.

BODE DIAGRAM

A Bode diagram consists of two graphs. One is the plot of the logarithm of the magnitude of a sinusoidal transfer function against the frequency on a logarithmic scale. The other is a plot of the phase angle against the frequency on a logarithmic scale.

The standard representation of the logarithmic magnitude of $G(j\omega)$ is $20\ \log_{10}|G(j\omega)|\ dB$. The Bode diagram uses semi-log graph sheet: a linear scale on the y-axis to represent the magnitude in decibels or phase angle in degrees, and a logarithmic scale on the x-axis to represent the frequency range of interest. The logarithmic scale allows to consider the characteristics at very low frequencies.

The main advantage in using the logarithmic plot is the relative ease of plotting the frequency response curves. When we consider the transfer function of $G(s)H(s)\ or\ G(j\omega)H(j\omega)$, the following factors occur frequently.

1. Gain K	2. Integral term $\frac{1}{s}$ and derivative factors s
3. First order factor $(s+a)\ or\ (1+Ts)$ in numerator or denominator	4. Quadratic factors $(s^2 + 2\zeta\omega_n s + {\omega_n}^2)$ or $\left(\frac{s}{{\omega_n}^2}^2 + \frac{2\zeta}{\omega_n}s + 1\right)$ in numerator or denominator

Gain K

Since K is a constant greater than zero,

$$20\ \log_{10}|G(j\omega)| = 20\ \log_{10} K\ dB$$

which is a constant irrespective of the value of ω

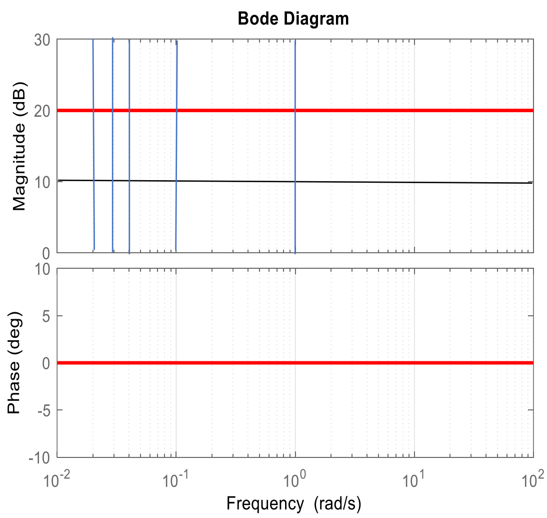
The magnitude plot will be a horizontal line.

$$< G(j\omega) = \tan^{-1} 0 = 0$$

The phase plot will be also a horizontal line at zero.

The figure shows the Bode diagram for $K=10$.

$$G(s) = 10$$



Integral factor $\left(\frac{1}{s}\right)$ (Pole at origin)

$$G(s) = \frac{1}{s} \quad \gg \quad G(j\omega) = \frac{1}{j\omega}$$

$$|G(j\omega)| = \left|\frac{1}{j\omega}\right| = \left|\frac{j\omega}{-\omega^2}\right| = \left|\frac{-j}{\omega}\right| = \frac{1}{\omega}$$

Magnitude plot :

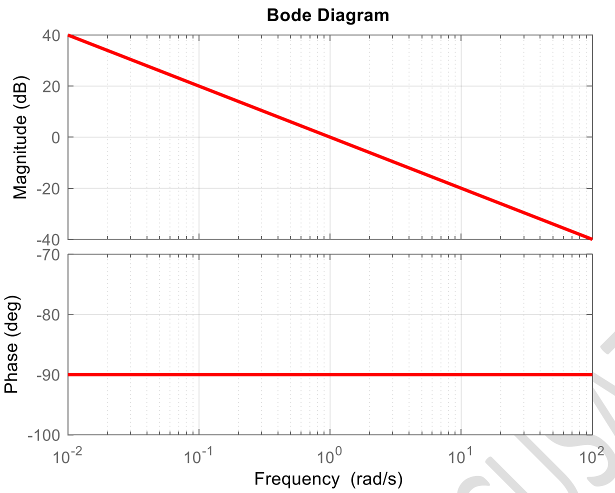
$$20\ \log_{10}|G(j\omega)| = 20\ \log_{10}\left(\frac{1}{\omega}\right)\ dB$$

$$= -20\log_{10}(\omega)\ dB$$

Slope of the line is $= -20\ dB/decade$

Phase plot

$$< G(j\omega) = \tan^{-1}\frac{-\frac{1}{\omega}}{0} = -90^0$$



Derivative factor (s) (Zero at origin)

$$G(s) = s \gg \quad G(j\omega) = j\omega$$

$$|G(j\omega)| = |j\omega| = \omega$$

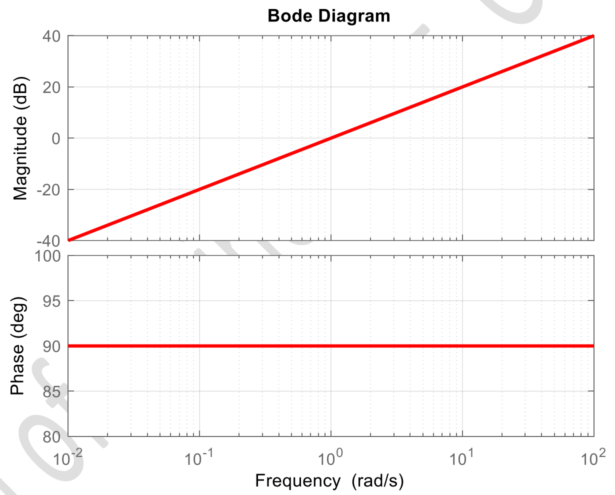
Magnitude plot :

$$20\ \log_{10}|G(j\omega)| = 20\ \log_{10}(\omega)\ dB$$

Slope of the line is $= 20\ dB/decade$

Phase plot

$$< G(j\omega) = \tan^{-1}\frac{\omega}{0} = +90^0$$



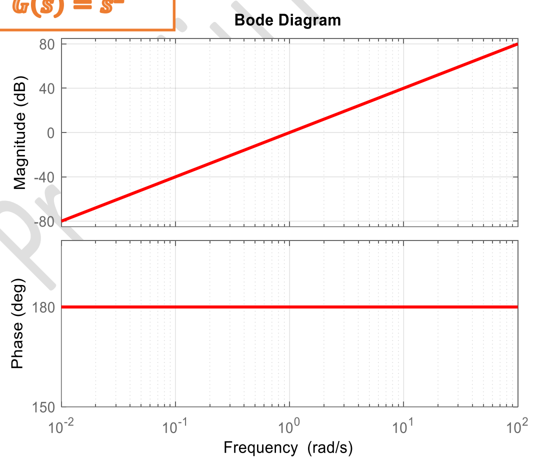
If we have $G(s) = s^n \quad \ggg \quad G(j\omega) = (j\omega)^n \quad \gg\gg \quad |G(j\omega)| = |\ (j\omega)^n| = \ \omega^n$

Magnitude plot : $\quad 20\ \log_{10}|G(j\omega)| = 20\ \log_{10}(\ \omega^n) = 20n\ \log_{10}(\omega)\ dB$

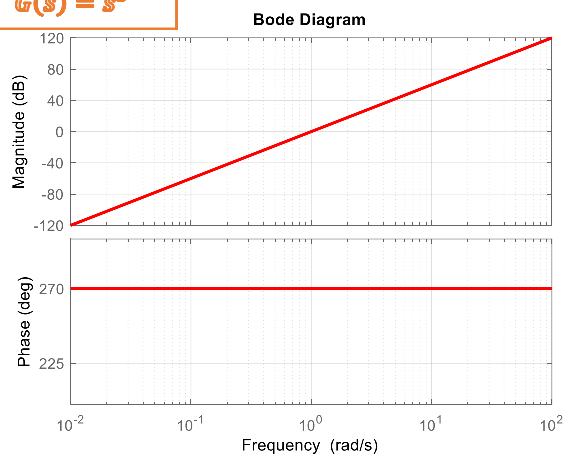
Slope of the line is $= 20n\ dB/decade$

Phase plot: $\quad < G(j\omega) = \ < (j\omega)^n = n\ < (j\omega) = +n.90^0$

$\boxed{G(s) = s^2}$



$\boxed{G(s) = s^3}$



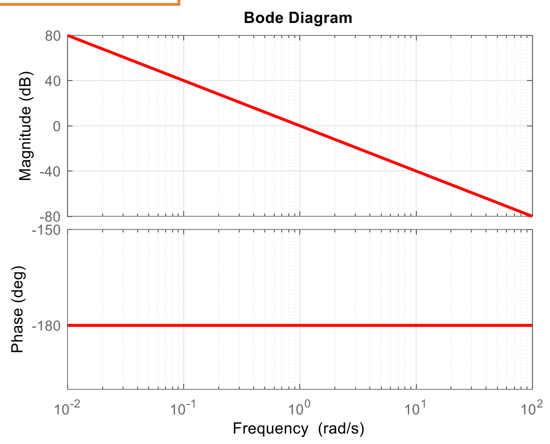
If we have $G(s) = \frac{1}{s^n}$ >>> $G(j\omega) = \left(\frac{1}{j\omega}\right)^n = \left(\frac{-j}{\omega}\right)^n$ >>>> $|G(j\omega)| = \left|\left(\frac{-j}{\omega}\right)^n\right| = \frac{1}{\omega^n}$

Magnitude plot : $20\ \log_{10}|G(j\omega)| = 20\ \log_{10}\left(\frac{1}{\omega^n}\right) = -20n\ \log_{10}(\omega)\ dB$

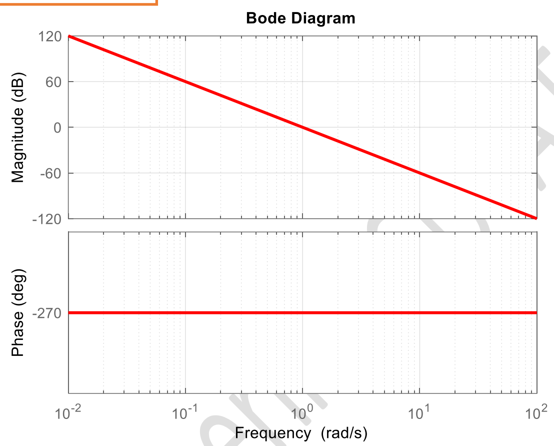
Slope of the line is $= -20n\ dB/decade$

Phase plot: $< G(j\omega) =< \left(\frac{-j}{\omega}\right)^n = n\ < \left(\frac{-j}{\omega}\right) = -n.90^0$

$G(s) = \frac{1}{s^2}$



$G(s) = \frac{1}{s^3}$



First order term $(s+a)$ or $(1+Ts)$ in the denominator (Pole on the real axis)

$$G(s) = \frac{a}{s+a} = \frac{1}{1+Ts}$$

$$G(j\omega) = \frac{1}{1+j\omega T} = \frac{1-j\omega T}{1+\omega^2T^2} = \frac{1}{1+\omega^2T^2} - j\frac{\omega T}{1+\omega^2T^2}$$

$$|G(j\omega)| = \left|\frac{1}{1+\omega^2T^2} - j\frac{\omega T}{1+\omega^2T^2}\right| = \frac{1}{\sqrt{1+\omega^2T^2}}$$

Magnitude plot :

$$20\ \log_{10}|G(j\omega)| = 20\ \log_{10}(1+\omega^2T^2)^{\frac{-1}{2}}\ dB$$
$$= -10\log_{10}(1+\omega^2T^2)\ dB$$

If $\omega T \ll 1,\quad 1+\omega^2T^2 \cong 1$ then
$20\ \log_{10}|G(j\omega)| = 0$

$\omega T \gg 1,\quad 1+\omega^2T^2 \cong (\omega T)^2,$

then $20\ \log_{10}|G(j\omega)| = -10\log_{10}(\omega T)^2 = -20\log_{10}\omega T$

When $\omega T = 10,\ 20\ \log_{10}|G(j\omega)| = -20$

When $\omega T = 100,\ 20\ \log_{10}|G(j\omega)| = -40$

When $\omega T = 1000,\ 20\ \log_{10}|G(j\omega)| = -60$

Slope of the line is $= -20\ dB/decade$

Phase plot

$$\angle G(j\omega) = \tan^{-1}(-\omega T)$$

If $\omega T \ll 1,\quad \tan^{-1}(-\omega T) = 0$

When $\omega T = 1,\ \tan^{-1}(-\omega T) = -45^0$

If $\omega T \gg 1,\quad \tan^{-1}(-\omega T) = -90^0$

The frequency $\omega = \frac{1}{T}$ is known as corner frequency or break frequency.

Example

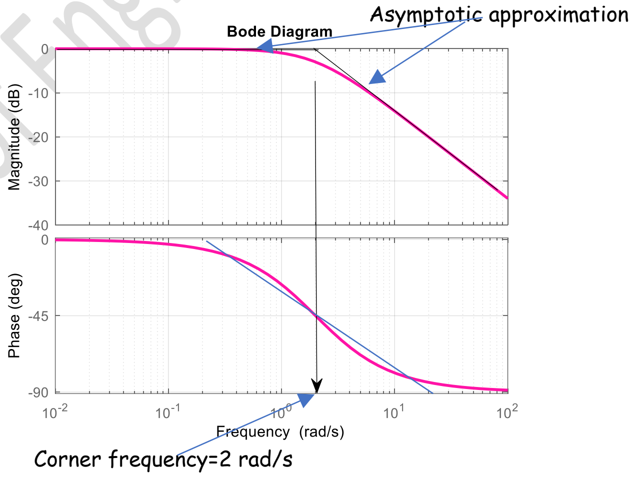
$$G(s) = \frac{2}{s+2} = \frac{1}{1+0.5s}$$

Corner frequency $\omega = \frac{1}{T} = \frac{1}{0.5} = 2\ rad/s$

$M = -10\log_{10}(1+\omega^2T^2),\qquad \theta = -\tan^{-1}(\omega T)$

ω	ωT	Mag(dB)	Phase(deg)
0.01	0.005	-0.0001	-0.286
0.1	0.05	-0.011	-2.86
1	0.5	-0.969	-26.56
2	1	-3.010	-45
10	5	-14.15	-78.69
100	50	-33.98	-88.85

We can also draw the asymptotes as shown to get an approximate magnitude plot.



First order term $(s+a)$ or $(1+Ts)$ in the numerator (Zero on the real axis)

$$G(s) = s + a = 1 + Ts$$
$$G(j\omega) = 1 + j\omega T$$
$$|G(j\omega)| = |1 + j\omega T| = \sqrt{1 + \omega^2 T^2}$$

Magnitude plot :

$$20\ \log_{10}|G(j\omega)| = 20\ \log_{10}(1+\omega^2T^2)^{\frac{1}{2}}\ dB$$
$$= 10\log_{10}(1+\omega^2T^2)\ \ dB$$

If $\omega T \ll 1$, $1+\omega^2T^2 \cong 1$ then $20\ \log_{10}|G(j\omega)| = 0$

$\omega T \gg 1$, $1+\omega^2T^2 \cong (\omega T)^2$,

then $20\ \log_{10}|G(j\omega)| = 10\log_{10}(\omega T)^2 = 20\log_{10}\omega T$

When $\omega T = 10$, $20\ \log_{10}|G(j\omega)| = 20$

When $\omega T = 100$, $20\ \log_{10}|G(j\omega)| = 40$

When $\omega T = 1000$, $20\ \log_{10}|G(j\omega)| = 60$

Slope of the line is $= 20\ dB/decade$

Phase plot

$$< G(j\omega) = \tan^{-1}(\omega T)$$

If $\omega T \ll 1$, $\tan^{-1}(\omega T) = 0$

When $\omega T = 1$, $\tan^{-1}(\omega T) = 45^0$

If $\omega T \gg 1$, $\tan^{-1}(\omega T) = 90^0$

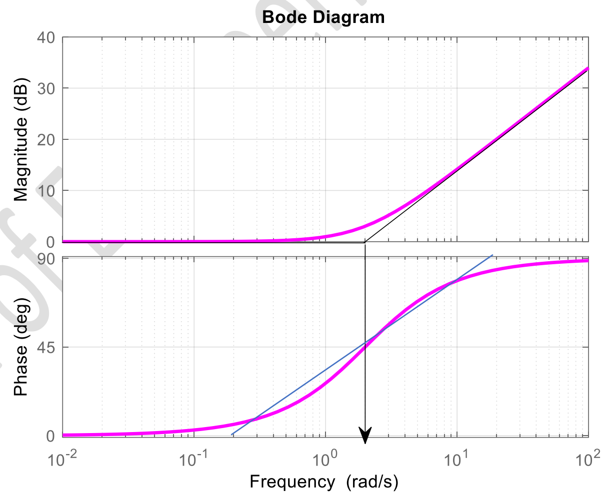
Example

$$G(s) = s + 2 = 1 + 0.5s$$

Corner frequency $\omega = \frac{1}{T} = \frac{1}{0.5} = 2\ rad/s$

$M = 10\log_{10}(1+\omega^2T^2)$, $\theta = \tan^{-1}(\omega T)$

ω	ωT	Mag(dB)	Phase(deg)
0.01	0.005	0.0001	0.286
0.1	0.05	0.011	2.86
1	0.5	0.969	26.56
2	1	3.010	45
10	5	14.15	78.69
100	50	33.98	88.85



A quadratic term $\frac{s}{\omega_n}^2 + \frac{2\zeta}{\omega_n}s + 1$ on the numerator (Complex conjugate zeros)

$$G(s) = \frac{s}{\omega_n}^2 + \frac{2\zeta}{\omega_n}s + 1$$

$$G(j\omega) = 1 + j\frac{2\omega\zeta}{\omega_n} + \frac{(j\omega)^2}{\omega_n{}^2} = 1 + j2\zeta\frac{\omega}{\omega_n} - \frac{\omega}{\omega_n}^2 = (1 - r^2) + j\,2r\zeta \quad where\ r = \frac{\omega}{\omega_n}$$

$$|G(j\omega)| = |(1 - r^2) + j\,2r\zeta\,| = \sqrt{(1 - r^2)^2 + (2r\zeta)^2}$$

Magnitude plot :

$$20\ \log_{10}|G(j\omega)| = 20\ \log_{10}((1 - r^2)^2 + (2r\zeta)^2)^{\frac{1}{2}}\ dB$$
$$= 10\log_{10}((1 - r^2)^2 + (2r\zeta)^2)\ dB$$

If $r \ll 1$, then $20\ \log_{10}|G(j\omega)| = 0$

If $r \gg 1$, $20\ \log_{10}|G(j\omega)| \cong 10\ \log_{10} r^4 = 40\log_{10} r$

When $r = 10$, $20\ \log_{10}|G(j\omega)| = 40$

When $r = 100$, $20\ \log_{10}|G(j\omega)| = 80$

When $r = 1000$, $20\ \log_{10}|G(j\omega)| = 120$

Slope of the line is $= 40\ dB/decade$

Phase plot

$$< G(j\omega) = \tan^{-1}\left(\frac{2r\zeta}{1 - r^2}\right)$$

If $r \ll 1$, $\tan^{-1}\left(\frac{2r\zeta}{1-r^2}\right) \cong 0$

When $r = 1$, $\tan^{-1}\left(\frac{2r\zeta}{1-r^2}\right) = 90^0$

If $r \gg 1$, $\tan^{-1}\left(\frac{2r\zeta}{1-r^2}\right) = 180^0$

Example

$$G(s) = \frac{1}{16}(s^2 + 4s + 16) = (1 + 0.25s + 0.0625s^2)$$

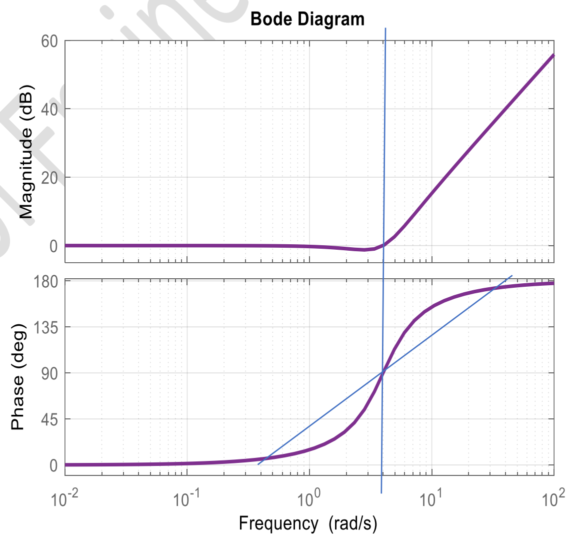
$$\omega_n = 4,\ \ \zeta = 0.5$$

Magnitude plot :

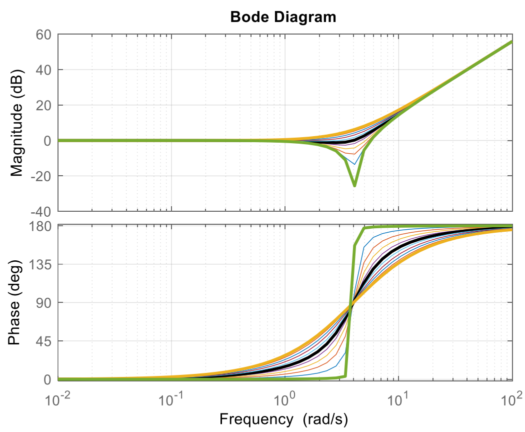
$$20\ \log_{10}|G(j\omega)| = 10\log_{10}[(1 - r^2)^2 + (2r\zeta)^2]\ dB$$

Phase plot : $< G(j\omega) = \tan^{-1}\left(\frac{2r\zeta}{1-r^2}\right)$

r	$\omega = 4r$	Mag(dB)	Phase(deg)
0.01	0.04	-0.00043	0.572
0.1	0.4	-0.0432	5.76
1	4	0	90
10	40	40	174.23
100	400	80	179.42



The plots can be approxiimated by asymptotic asuumptions. But around $r = 1$, the exact shape depends on the value of damping factor ζ. The above example is solved for values of ζ ranging from 0.1 to 1.



A quadratic term $\frac{s}{\omega_n}^2 + \frac{2\zeta}{\omega_n}s + 1$ on the denominator (Complex conjugate poles)

$$G(s) = \frac{1}{\frac{s}{\omega_n}^2 + \frac{2\zeta}{\omega_n}s + 1}$$

$$G(j\omega) = \frac{1}{1 + j\frac{2\omega\zeta}{\omega_n} + \frac{(j\omega)^2}{\omega_n^2}} = \frac{1}{(1-r^2) + j\,2r\zeta} = \frac{(1-r^2) - j\,2r\zeta}{(1-r^2)^2 + (\,2r\zeta)^2} = \frac{(1-r^2)}{(1-r^2)^2 + (\,2r\zeta)^2} - j\frac{2r\zeta}{(1-r^2)^2 + (\,2r\zeta)^2}$$

$$|G(j\omega)| = \frac{\sqrt{(1-r^2)^2 + (2r\zeta)^2}}{(1-r^2)^2 + (\,2r\zeta)^2} = \frac{1}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}}$$

Magnitude plot :

$$20\ \log_{10}|G(j\omega)| = 20\ \log_{10}((1-r^2)^2 + (2r\zeta)^2)^{\frac{-1}{2}}\ dB$$
$$= -10\log_{10}((1-r^2)^2 + (2r\zeta)^2)\ dB$$

If $r \ll 1$, then $20\ \log_{10}|G(j\omega)| = 0$

If $r \gg 1$, $20\ \log_{10}|G(j\omega)| \cong -10\ \log_{10} r^4 = -40\log_{10} r$

When $r = 10$, $20\ \log_{10}|G(j\omega)| = -40$

When $r = 100$, $20\ \log_{10}|G(j\omega)| = -80$

When $r = 1000$, $20\ \log_{10}|G(j\omega)| = -120$

Slope of the line is $= -40\ dB/decade$

Phase plot

$$< G(j\omega) = -\tan^{-1}\left(\frac{2r\zeta}{1-r^2}\right)$$

If $r \ll 1$, $\tan^{-1}\left(-\frac{2r\zeta}{1-r^2}\right) \cong 0$

When $r = 1$, $\tan^{-1}\left(-\frac{2r\zeta}{1-r^2}\right) = -90^0$

If $r \gg 1$, $\tan^{-1}\left(-\frac{2r\zeta}{1-r^2}\right) = -180^0$

Example

$$G(s) = \frac{16}{(s^2 + 4s + 16)} = \frac{1}{1 + 0.25s + 0.0625s^2}$$

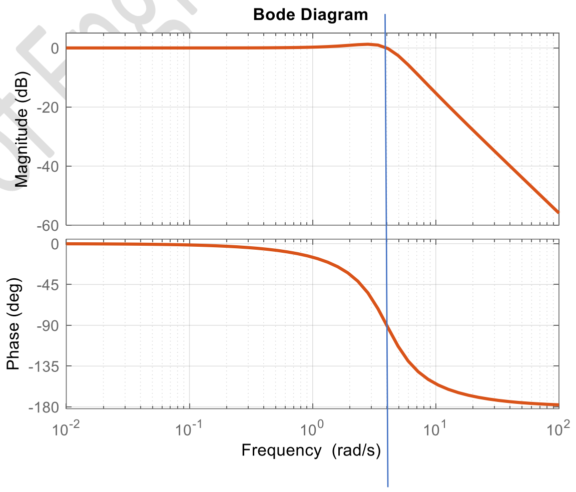
$$\omega_n = 4,\ \ \zeta = 0.5$$

Magnitude plot :

$20\ \log_{10}|G(j\omega)| = -10\log_{10}[(1-r^2)^2 + (2r\zeta)^2]\ dB$

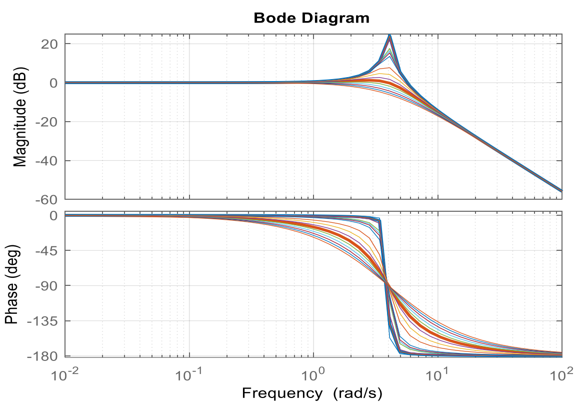
Phase plot : $< G(j\omega) = -\tan^{-1}\left(\frac{2r\zeta}{1-r^2}\right)$

r	$\omega = 4r$	Mag(dB)	Phase(deg)
0.01	0.04	0.00043	-0.572
0.1	0.4	0.0432	-5.76
1	4	0	-90
10	40	-40	-174.23
100	400	-80	-179.42



The plots can be approximated by asymptotic asuumptions. But around $r = 1$, the exact shape depends on the value of damping factor ζ. The above example is solved for values of ζ ranging from 0.1 to 1.

If the damping ratio ζ is zero, the magnitude of $G(j\omega)$ approaches ∞ as $r \to 1$. This is the resonant condition. As ζ varies from 0 to 0.707, the magnitude is having a peak at a frequency ratio $r = \sqrt{1-2\zeta^2}$. The maximum value of the magnitude $M_p = \frac{1}{2\zeta\sqrt{1-2\zeta^2}}$



The above results are obtained by differentiating $|G(j\omega)| = \frac{1}{\sqrt{(1-r^2)^2+(2r\zeta)^2}}$ with respect to r and equating it to zero.

Example

Draw the Bode diagram for a unity negative feedback system having a forward transfer function $G(s) = \frac{K(s+3)}{s(s+1)(s+2)}$ by taking $K = 1$.

$G(s) = \frac{(s+3)}{s(s+1)(s+2)} = \frac{\frac{3}{2}(1+\frac{s}{3})}{s(1+s)(1+\frac{s}{2})}$: $G(j\omega) = \frac{\frac{3}{2}(1+\frac{j\omega}{3})}{(j\omega)(1+j\omega)(1+\frac{j\omega}{2})}$

The corner frequencies are $1, 2, and\ 3$. We have

1. A constant $\frac{3}{2}$ in the numerator	2. A first order term in the numerator (zero at $s = -3$)
3. An integrating factor (pole at the origin, $s = 0$)	4. A first order term in the denomiator (pole at $s = -1$)
5. A first order term in the denomiator (pole at $s = -2$)	

Magnitude plot:

$$20\ \log_{10}|G(j\omega)| = 20\ \log_{10}\left|\frac{\frac{3}{2}(1+\frac{j\omega}{3})}{(j\omega)(1+j\omega)(1+\frac{j\omega}{2})}\right| = 20\ \log_{10}\left\{\frac{|1.5|\,|1+\frac{j\omega}{3}|}{|j\omega||1+j\omega||1+\frac{j\omega}{2}|}\right\}$$

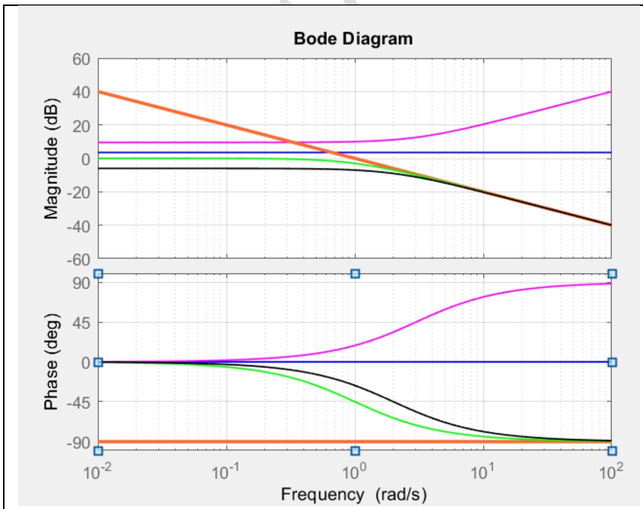
$$= 20\ \log_{10}\left|\frac{3}{2}\right| + 20\ \log_{10}\left|1+\frac{j\omega}{3}\right| - 20\ \log_{10}|j\omega| - 20\ \log_{10}|1+j\omega| - 20\ \log_{10}\left|1+\frac{j\omega}{2}\right|$$

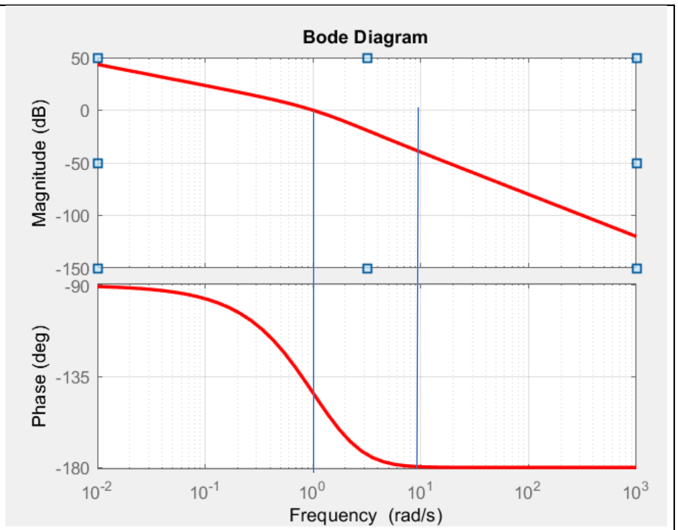
Phase Plot:

$$< G(j\omega) =< \left[\frac{\frac{3}{2}(1+\frac{j\omega}{3})}{(j\omega)(1+j\omega)(1+\frac{j\omega}{2})}\right] =< \left(\frac{3}{2}\right) +< \left(1+\frac{j\omega}{3}\right) -< (j\omega) -< (1+j\omega) -< \left(1+\frac{j\omega}{2}\right)$$

	1.5		$(1+j0.33\omega)$ $T = 0.33$		$(j\omega)^{-1}$		$(1+j\omega)^{-1}$ $T = 1$		$(1+j0.5\omega)^{-1}$ $T = 0.5$		Σ	
ω	Mag(dB) $20\ \log_{10} K$	Phase(deg) $\tan^{-1} 0$	Mag $10\log(1+\omega^2T^2)$	Phase $\tan^{-1}(\omega T)$	Mag $-20\log(\omega)$	Phase $\tan^{-1}\frac{-\frac{1}{\omega}}{0}$	Mag $-10\log(1+\omega^2T^2)$	Phase $\tan^{-1}(-\omega T)$	Mag $-10\log(1+\omega^2T^2)$	Phase $\tan^{-1}(-\omega T)$	Mag dB	Phase deg
0.01	3.52	0	0.00005	0.19	40	-90	-0.0004	-0.573	-0.0001	-0.286	43.5	-90.
0.1	3.52	0	0.0048	1.91	20	-90	-0.043	-5.71	-0.0108	-2.86	23.5	-97
1	3.52	0	0.458	18.44	0	-90	-3.010	-45	-0.969	-26.56	0	-143
2	3.52	0	1.6	33.69	-6.02	-90	-6.989	-63.43	-3.010	-45	-11	-164
3	3.52	0	3	45	-9.54	-90	-10	-71.56	-5.118	-56.31	-18	-173
10	3.52	0	10.83	73.3	-20	-90	-20	-84.28	-14.15	-78.69	-40	- 179.7
100	3.52	0	30.45	88.28	-40	-90	-40	-89.42	-33.98	-88.85	-80	-180
1000	3.52	0	50.45	89.8	-60	-90	-60	-89.94	-53.98	--89.89	-120	-180

The plots of individual factors and that of combined transfer function are shown below.





Example:

Draw the Bode diagram for the following transfer function $G(s) = \frac{10(s+3)}{s(s+2)(s^2+s+2)}$

The quadratic factor in the denominator gives $\omega_n = 1.414 \frac{rad}{s}$ $and\ \zeta = 0.353$

$$G(s) = \frac{10 \times 3(1+\frac{s}{3})}{2\times 2\times s(1+\frac{s}{2})(1+\frac{s}{2}+\frac{s^2}{2})} \quad : \quad G(j\omega) = \frac{7.5(1+\frac{j\omega}{3})}{(j\omega)(1+\frac{j\omega}{2})\left[1+\frac{j\omega}{2}+\frac{(j\omega)^2}{2}\right]}$$

Corner frequencies are $1.414, 2, 3\ rad/s$

Magnitude plot:

$$20\ \log_{10}|G(j\omega)| = 20\ \log_{10}\left|\frac{7.5(1+\frac{j\omega}{3})}{(j\omega)(1+\frac{j\omega}{2})\left[1+\frac{j\omega}{2}+\frac{(j\omega)^2}{2}\right]}\right|$$

$$= 20\ \log_{10}|7.5| + 20\ \log_{10}\left|1+\frac{j\omega}{3}\right| - 20\ \log_{10}|j\omega| - 20\ \log_{10}\left|1+\frac{j\omega}{2}\right| - 20\ \log_{10}\left|1+\frac{j\omega}{2}+\frac{(j\omega)^2}{2}\right|$$

Phase Plot:

$$< G(j\omega) =< \left[\frac{7.5(1+\frac{j\omega}{3})}{(j\omega)(1+\frac{j\omega}{2})\left[1+\frac{j\omega}{2}+\frac{(j\omega)^2}{2}\right]}\right] =< (7.5) +< \left(1+\frac{j\omega}{3}\right) -< (j\omega) -< \left(1+\frac{j\omega}{2}\right) -< \left(1+\frac{j\omega}{2}+\frac{(j\omega)^2}{2}\right)$$

	7.5		$(1+j0.33\omega)$ $T=0.33$		$(j\omega)^{-1}$		$(1+j0.5\omega)^{-1}$ $T=0.5$		$\left(1+\frac{j\omega}{2}+\frac{(j\omega)^2}{2}\right)^{-1}$ $\omega_n = 1.414\ and\ \zeta = 0.353$			Σ	
ω	Mag $20\ \log_{10} K$	Phase $\tan^{-1} 0$	Mag $10\log(1+\omega^2T^2)$	Phase $\tan^{-1}(\omega T)$	Mag $-20\log(\omega)$	Phase $\tan^{-1}\frac{-\frac{1}{\omega}}{0}$	Mag $-10\log(1+\omega^2T^2)$	Phase $\tan^{-1}(-\omega T)$	$-10\log\ (1-r^2)^2+(2r\zeta)^2$		$-\tan^{-1}\left(\frac{2r\zeta}{1-r^2}\right)$	Mag dB	Phase deg
0.01	17.5	0	0.00005	0.19	40	-90	-0.0001	-0.286	r=0.007	0.0003	-0.286	57.5	-90
0.1	17.5	0	0.0048	1.91	20	-90	-0.0108	-2.86	r=0.07	0.0326	-2.87	37.52	-93.8
1	17.5	0	0.458	18.44	0	-90	-0.969	-26.56	r=0.707	0.302	-45	17.29	-143.
1.414	17.5	0	0.871	25.24	-3.01	-90	-1.76	-35.26	r=1	3.02	-90	16.62	-190
2	17.5	0	1.6	33.69	-6.02	-90	-3.010	-45	r=1.414	-3	-135	7.08	-236
3	17.5	0	3	45	-9.54	-90	-5.118	-56.31	r=2.12	-11.6	-156.8	-5.8	-258
10	17.5	0	10.83	73.3	-20	-90	-14.15	-78.69	r=7.07	-33.8	-174.18	-39.15	-269
100	17.5	0	30.45	88.28	-40	-90	-33.98	-88.85	r=70.7	-73.8	-179.43	-100	-270
1000	17.5	0	50.45	89.8	-60	-90	-53.98	--89.89	r=707	-113.8	-179.95	-160	270